MILITARY SPECIFICATION

PROJECTILE, 155MM, ILLUMINATING M485A2, METAL PARTS ASSEMBLY

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers, Projectile, Illuminating, M485A2, Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-M-6867	-	Magnetic Inspection Units
MIL-I-6868	-	Inspection Process, Magnetic Particle
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	-	Certification of Magnetic Particle Inspection Personnel
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures for Sampling by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

7549275	-	Pallet for 155MM Projectile
9214142	-	Projectile, 155MM, Illuminating, M485A2 Metal Parts Assembly

INSPECTION EQUIPMENT DRAWINGS

9280363	-	Magnetic Particle Standard for M485A2 Body
9280364	-	Magnetic Particle Standard for Base Plug, 155MM, M485A2

PUBLICATIONS

ARDC

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

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ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

ASTM Method D638-77A	-	Tensile Properties of Plastics
ASTM Method E-8	-	Tension Testing and Metallic Materials
ASTM Method E-10	-	Brinell Hardness of Metallic Materials
ASTM Method E103	-	Rapid Indentation Hardness Testing of Metallic Materials
ASTM Method B117	-	Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Heat treatment. Each projectile body and each base plug shall be austenitized, oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawings. No work involving plastic flow is permitted after tempering. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees Fahrenheit (F), suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees Fahrenheit (F), after which cooling may be accelerated by water spray.

3.3 Projectile body. The minimum (min.) mechanical properties as required by the applicable drawing shall apply to specimens machined from any part of the projectile body.

3.4 Base plug. The min. mechanical properties required by the applicable drawing shall apply to specimens machined from the base plug.

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3.5 Rotating band.

3.5.1 Seat. Prior to application of the rotating band blank, the seat shall be clean and free of oil, grease, dirt, rust and other foreign material. Shot or sandblasting shall not be permitted for cleaning the band seat after scoring.

3.5.2 Band blank - (gilding metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification, except that when hot banding at not less than 800°F, no prior annealing is necessary.

3.5.3 Application and tightness. The rotating band blank shall be carefully pressed into the band seat after final heat treatment of the projectile body so as to fit tightly with no distortion of the adjacent walls. It shall make contact with the sidewals of the band seat for a min. of 50 percent (%) of the depth of the seat and for a full 360 degrees (°) at the bourrelet, when tested as specified in 4.4.3.2. There shall be no gap between the rotating band and the sidewalls of the band seat after the rotating band has been machined to final dimension.

3.5.3.1 Applied temperature. The bands may be applied either hot or cold, but if applied hot, the bands shall not be heated above 1600° Fahrenheit (F) to prepare them for application to the body.

3.5.3.2 Heat conditioning. If the bands are heated for application, the bands shall be heated uniformly by a process which will prevent undue oxidation (loose scale), and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.5.4 Seating. The diametrical clearance between the rotating band and band seat shall not exceed .006 inches when the destructive method of testing is used, or .009 inches when the non-destructive method of testing is used.

3.5.5 Mechanical properties. After application to the projectile, and subsequent removal, the rotating band shall have a min. tensile strength of 30,000 pounds per square inch (psi) and a min. elongation of 25% of the gage length when tested as specified in 4.5.2.4.

3.6 Metal defects.

3.6.1 Projectile body and base plug. The projectile body and base plug shall be free from all metal defects, in accordance with the defect standards (see 3.6.2) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of each component shall

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be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits (see 3.6.2)

3.6.2 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first months production or whenever requested by the procuring contracting agency. The establishment of these standards however, will not waive the requirement for compliance with 3.6.1.

3.7 Obturating band.

3.7.1 Band defects. The obturating band shall be uniform in color, free of cracks, burn marks, flashing, contamination and shall have a smooth finish. Surface irregularities resulting from cutting off spurs and gates shall be removed by grinding and buffing until no protrusions exist.

3.7.2 Band strength. When subjected to the test, the pull strength and elongation shall be 325 pounds min. and 4.5% min., respectively.

3.7.3 Moisture content of obturating band inspection sample. The obturating band shall be moisture-conditioned by immersion in boiling water until 2.0 to 3.0%, be weight, moisture content is attained.

3.8 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Proving ground functional test. The projectile assembly shall reveal no evidence of failure when ballistically tested at Excess and Service pressures (see 4.5.7). Metal part failures are defined in 3.9.1 and 3.9.2.

3.9.1 Excess pressure. There shall be no crack, breakup, deformation or evidence of failure of the projectile assembly in bore of the weapon or in flight.

3.9.2 Service pressure (fired for recovery).

3.9.2.1 Deformation occurring in the bore of the gun as evidenced by rifling marks extending more than 180° around the body or by an increase or decrease in the diameter of the projectile body at the bourrelet, and just forward of and to the rear of the rotating band.

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3.9.2.2 Shear of the rotating band as evidenced by a complete lack of engraving.

3.9.2.3 Dishing or deformation of the base.

3.9.2.4 Projectile cracked or broken by firing.

3.10 Protective finish. Details of the proposed procedure for zinc phosphate coating to be used by the contractor shall be submitted in writing to the Government Technical Agency and written approval received prior to initiation of production. The proposed procedures shall include materials, processes and equipment to be used by the contractor, detailed methods of control including limits for time, temperatures concentration and all other information pertinent to the processes. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Burr and sharp edges. No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.11.2 Foreign matter. All parts and assemblies shall be free of chips, dirt, greast, oil, rust, or other foreign matter which may interfere with the assembly, loading, function or final inspection and be completely free of same prior to shipping.

3.11.3 Threads. Threads shall be full and undamaged, for the entire minimum length or depth as required by the applicable drawings.

3.11.4 Painting. Painting shall comply with the requirements of the applicable drawing, complete, void free and undamaged. Drying time prior to testing shall be in accordance with applicable specification or drawing.

3.11.5 Finish. Exterior surface coatings shall be continuous and without blemishes. All markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

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a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated:

PART DESCRIPTION	DRAWINGS	QUANTITY
Projectile Metal Parts Assembly Complete with Finished Base Plugs (unassembled)	9214142	20
Body Assembly, Complete	9362659	4
Body, Projectile, Prior to Machining	9214143	4
Body, Projectile, Machined, but Prior to Banding	9214143	4
Base Plug, Complete	9214146	4
Base Plug, Prior to Machining	9214146	3
Obturator Band	9214144	5
Rotating Band Blank	9214145	4

Twenty of the above projectile metal parts assemblies and bases shall be forwarded to a designated load plant for loading and shall then be forwarded to a Government proving ground for ballistic testing.

Five (5) obturating band molded samples made in accordance with ASTM-D638, Type 1 specimens with a thickness of 0.125 ± 0.015 to test the plastic material used to injection-mold or extrude the obturator bands. The specimens and obturator bands shall be moisture-conditioned and heat-sealed in plastic polyethylene bags to retain the moisture.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Projectile Metal Parts for Projectile, 155MM, Illuminating, M485A2		1 OF 1		SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Obturator Band (Dwg 9214144)				
	Examination for defects	5		3.1	4.4.2.1
	Base Plug (Dwg 9214146)				
	Examination for defects	4		3.1	4.4.2.2
	Yield strength & percent elongation	2		3.3	4.5.1.2
	Salt spray test	2		3.1	4.5.3/4.5.3.2
	Projectile Body (Dwg 9214143)				
	Examination for defects	4		3.1	4.4.2.3
	Yield strength & percent elongation	2		3.3	4.5.1.2
	Rotating Band Blank (Dwg 9214145				
	Examination for defects	5		3.1	4.4.2.4
	Body Assembly (Dwg 9362659				
	Examination for defects	4		3.1	4.4.2.5
	Salt spray test	2		3.1	4.5.3/4.5.3.1
	Projectile Metal Parts (Dwg 9214142)				
	Examination for defects	20		3.1	4.4.2.6
	Ballistic tests				4.5.7
	a. Excess pressure	15		3.9	4.5.7.1
	b. Service pressure	5		3.9	4.5.7.2

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Base plugs from not more than one forging manufacturer.

b. Projectile bodies from not more than one forging manufacturer.

4.4.1.1 Projectile body and base plug heat treatment lot. A heat treatment lot of bodies or bases is defined as follows:

4.4.1.1.1 Standard procedure. A heat treatment lot of projectile bodies or base plugs shall consist of bodies or bases from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process. The maximum lot size shall be 2000 bodies or 2000 bases. If a mill heat exceeds the maximum lot size, the mill heat shall be divided into any convienent number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots.

4.4.1.1.1.1 Miscellaneous lot. Projectile bodies or base plugs manufactured from small heats of steel, or bodies/bases that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1.1. The maximum lot size shall be 2000 bodies or 2000 bases.

4.4.1.1.2 Use. The heat treatment lot constituted per 4.4.1.1.1, 4.4.1.1.1.1 shall be the basis for selecting samples for mechanical properties tests. Projectile bodies or base plugs in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Projectile lot. Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination. See MIL-A-48078.

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a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize capability in accordance with contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214144
4.4.2.1	Obturator Band				NEXT HIGHER ASSEMBLY 9214142
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRTICIAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Width		0.40%	3.1	Gage
103	Thickness		0.40%	3.1	Gage
104	Total length		0.40%	3.1	Gage
105	Moisture content		*	3.7.3	4.4.3.3.1/4.5.4
106	Strength test		*	3.7.2	4.4.3.3.2/4.5.5
MINOR					
201	Diameter of hole (2 holes, 4 places)		0.65%	3.1	Gage
202	Location of hole (4 places)		0.65%	3.1	Gage
203	Radii or chamfers missing or incorrect		0.65%	3.1	Gage
204	Protrusion depth (if applicable)		0.65%	3.1	Gage
205	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

* Sampling size as specified in applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Base Plug		9214146
			NEXT HIGHER ASSEMBLY 9214142

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRTICIAL					
1	Metal defective		100%	3.6.1	4.5.6
MAJOR					
101	Diameter of flange		0.40%	3.1	Gage
102	Diameter of press fit area		0.40%	3.1	Gage
103	Length of press fit area		0.40%	3.1	Gage
104	Runout of outside diameter of flange with press fit diameter		0.40%	3.1	Gage
105	Underside of flange perpendicular with press fit diameter		0.40%	3.1	Gage
106	Yield strength and percent elongation	*		3.2	4.5.1.2
107	Protective coating (salt spray)	*		3.2	4.5.3
MINOR					
201	Total length		0.65%	3.1	Gage
202	Surface finish of press fit area		0.65%	3.1	Visual
203	Angle missing on flange		0.65%	3.1	Visual
204	Chamfer missing from leading edge of press fit area		0.65%	3.1	Visual
205	Depth of undercut, max.		0.65%	3.1	Gage
206	Width of undercut, max.		0.65%	3:1	Gage
207	Flat on flange, max.		0.65%	3.1	Visual
208	Protective finish inadequate or damaged		0.65%	3.1	Visual
209	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9214143
4.4.2.3	Projectile Body, Prior to Banding				NEXT HIGHER ASSEMBLY 9214142
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRTICIAL					
1	Metal defective		100%	3.6.1	4.5.6
MAJOR					
101	Pitch diameter of thread, max.		0.40%	3.1	Gage
102	Pitch diameter of thread, min.		0.40%	3.1	Gage
103	Minor diameter of thread, max.		0.40%	3.1	Gage
104	Length of effective thread, min.		0.40%	3.1	Gage
105	Runout of pitch diameter of thread with bourrelet and front face (2 places)		0.40%	3.1	Gage
106	Total length		0.40%	3.1	Gage
107	Runout of boattail with bourrelet		0.40%	3.1	Gage
108	Runout of inside diameter with bourrelet		0.40%	3.1	Gage
109	Length from rear face to expelling charge cavity		0.40%	3.1	Gage
110	Diameter of forward bourrelet		0.40%	3.1	Gage
111	Diameter of base plug cavity		0.40%	3.1	Gage
112	Length to first basic diameter of boattail		0.40%	3.1	Gage
113	Length to second basic diameter of boattail		0.40%	3.1	Gage
114	Inside diameter of large cavity		0.40%	3.1	Gage
115	Length from rear face to rear of rotating band seat		0.40%	3.1	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER 9214143
4.4.2.3	Projectile Body, Prior to Banding		NEXT HIGHER ASSEMBLY 9214142

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Diameter of rotating band seat		0.40%	3.1	Gage
117	Width of rotating band seat		0.40%	3.1	Gage
118	Diameter of obturating band seat		0.40%	3.1	Gage
119	Diameter of aft bourrelet		0.40%	3.1	Gage
120	Wall thickness of expelling charge cavity, min., max.		0.40%	3.1	Gage
121	Length of press fit area		0.40%	3.1	Gage
122	Scoring missing on band seat		0.40%	3.1	Visual
123	Wall thickness at forward end of large cavity, min.		0.40%	3.1	Gage
124	Runout of base plug diameter with bourrelet diameter		0.40%	3.1	Gage
125	Runout of obturator band seat with bourrelet diameter		0.40%	3.1	Gage
127	Runout of rotating band seat with bourrelet diameter		0.40%	3.1	Gage
128	Runout of external diameter rear of bourrelet with bourrelet diameter		0.40%	3.1	Gage
129	Foreign matter on rotating band seat		0.40%	3.5.1	Visual
130	Yield strength and percent elongation	*		3.2	4.5.1.2

NOTES:

* Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9214143
4.4.2.3	Projectile Body, Prior to Banding				NEXT HIGHER ASSEMBLY 9214142
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Diameter at front face of projectile, external		0.65%	3.1	Gage
202	Diameter of recess in expelling charge cavity		0.65%	3.1	Gage
203	Length from front face to recess in expelling charge cavity		0.65%	3.1	Gage
204	Width of obturating band seat		0.65%	3.1	Gage
205	Width between obturating band and rotating band seats		0.65%	3.1	Gage
206	Diameter of thread chamfer		0.65%	3.1	Gage
207	Width of chamfer at rear of projectile		0.65%	3.1	Gage
208	Surface finish of press fit area		0.65%	3.1	Gage
209	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
210	Surface finish improper		0.65%	3.1	Visual
211	Evidence of shot or sandblast in rotating band seat		0.65%	3.5.1	Visual
212	Evidence of poor workmanship		0.65%	3.11	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9214145
4.4.2.4	Rotating Band Blank		NEXT HIGHER ASSEMBLY 9214142

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
103	Width		0.40%	3.1	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.11	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9362659
4.4.2.5	Body Assembly (after painting)				NEXT HIGHER ASSEMBLY 9214142
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Runout of rotating band with bourrelet and internal base diameter		0.40%	3.1	Gage
102	Diameter of rotating band		0.40%	3.1	Gage
103	Diameter between rotating bands		0.40%	3.1	Gage
104	Length from rear face to rear shoulder of rotating band		0.40%	3.1	Gage
105	Diameter of bourrelet, max.		100%	3.1	Gage
106	Rotating band and bandseat clearance		*	3.5.3	4.4.3.2/4.5.2
107	Mechanical properties of rotating band		*	3.1	4.4.3.2.3/4.5.2.3
108	Salt spray		*	3.1	4.5.3
MINOR					
201	Width of rotating band		0.65%	3.1	Gage
202	Depth of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
203	Width of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
204	Radii or chamfer missing or incorrect		0.65%	3.1	Visual
205	Width of match between rotating band		0.65%	3.1	Gage
206	Protective coating inadequate or damaged		0.65%	3.1	Visual
207	Paint on threads		0.65%	3.11	Visual
208	Paint on base fit area		0.65%	3.11	Visual
209	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

* Sampling size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214142
4.4.2.6	Projectile Metal Parts				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
SPECIAL					
a	Ballistic test				4.5.7
	a. Excess pressure	*		3.9	4.5.7.1
	b. Service pressure	*		3.9	4.5.7.2
MAJOR					
101	Rotating band damaged		0.40%	3.1	Visual
102	Obturator missing or damaged		0.40%	3.1	Visual
103	Any component missing or improperly assembled		0.40%	3.1	Visual
104	Threads damaged		0.40%	3.11.3	Visual
105	Weight		0.40%	3.1	Scale
MINOR					
201	Marking missing, incorrect or illegible		0.65%	3.1	Visual
202	Burr		0.65%	3.11.1	Visual
203	Foreign matter		0.65%	3.11.2	Visual
204	Grommet missing or loose		0.65%	3.1	Visual
205	Corrosive preventive compound inadequate or excessive		0.65%	3.1	Visual
206	Lifting plug missing		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

* Sample size as specified in applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48043A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7549275
4.4.2.7	Pallet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Pallet damaged (unserviceable)		0.65%	3.1	Visual
202	Improperly packed		0.65%	3.1	Visual
203	Strapping missing, loose or broken		0.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Table and herein.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness (projectile body and base plug). Each projectile body and each base plug in their respective heat treat or heat treat miscellaneous lots shall be subject to a Brinnel Hardness Test or any other method aproved by the procuring activity. The hardness test shall be performed in accordance with 4.5.1.1.

4.4.3.1.2 Tensile tests (projectile body and base plug). Sample test quantities of projectile bodies and base plugs shall be subject to a yield strength and percent elongation test. These tests shall be performed in accordance with 4.5.1.2.

4.4.3.2 Rotating band.

4.4.3.2.1 Seating (non-destructive method). Five (5) body assemblies from each 1-hour production per banding machine in operation shall be subjected to the test specified in 4.5.2. Failure of one or more bands to comply with 3.5.4 shall be cause for rejection of the assemblies represented by the failing sample.

4.4.3.2.2 Seating (destructive method). One (1) body assembly shall be selected from each eight hours of production, or fraction thereof, from each banding machine in operation and tested as specified in 4.5.2. Failure of one or more bands to comply with 3.5.4 shall be cause for rejection of the assemblies represented by the failing sample.

4.4.3.2.3 Sidewall contact. Rotating bands removed as specified under 4.4.3.2.2 or 4.4.3.2.4 shall be examined as specified in 4.5.2.3 to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of assemblies represented by the sample.

4.4.3.2.4 Mechanical properties of rotating band. One (1) body assembly shall be selected from each lot produced during an 8-hour production or fraction thereof from each banding machine in operation and subjected to the test specified in 4.5.2.4 to determine compliance with 3.5.5. Failure to comply shall be cause for rejection of assemblies represented by the sample.

4.4.3.3 Obturating band.

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4.4.3.3.1 Moisture content of obturating band (see 3.7). Ten (10) obturating bands shall be selected from each lot and subjected to the test specified in 4.5.4. If two or more bands fail to comply with the applicable requirement, the lot shall be rejected. If one band fails to comply with the applicable requirements, an additional ten samples shall be selected. If in the combined sample two or more bands fail the requirements the lot shall be rejected.

4.4.3.3.2 Obturating band strength (see dwg. 9214144). Ten (10) obturating bands shall be selected from each lot and subjected to the test specified in 4.5.5. If two or more bands fail to comply with the requirements of 3.7.2 the lot shall be rejected. If one band fails to comply with the requirements of 3.7.2 an additional ten samples shall be selected for test and if three or more samples fail to comply with the requirements of 3.7.2 in the combined 20 samples tested, the lot shall be rejected.

4.4.4 Magnetic particle inspection (bodies and bases).

4.4.4.1 Projectile body. Heat treated projectile bodies after final machining but prior to banding shall be inspected by the magnetic particle inspection technique.

4.4.4.2 Base plugs. Heat treated base plugs after final machining but prior to applying protective coating shall be inspected by the magnetic particle inspection technique.

4.4.5 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification and/or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.5.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

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4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1. Hardness (projectile body and base plug). In order to determine uniformity of heat treatment, each projectile body and base plug after being heat treated and before final machining, shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the projectile body three and one quarter (3 1/4) inches forward of the finished rear face of the body in the area of the rotating band. This location must be approximately increased to allow for the extra length of a rough turned body. Readings taken on the base plug shall be on the longitudinal centerline of the base plug. The test shall be made in accordance with ASTM method E-103 and the following:

a. The surfaces of the projectile body and base plug in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, 0.015 inches minimum, to remove scale and get below the decarburization zone. (Note: Flats and impressions shall be completely removed by final machining.)

b. Hardness tester periodic verification shall be performed using brinnel test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded (displayed).

d. The hardness tester shall use a standard 10mm brinnel ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.1.2 Yield strength and percent elongation of projectile body and base plug. The hardest and two (2) softest bodies or bases from each standard heat treatment lot or the two (2) hardest and four (4) softest bodies or bases from each miscellaneous heat treatment lot

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as determined by the hardness test in 4.5.1.1 shall be subjected to this test. Sample for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. For the projectile body two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the band seat area and two specimens as prescribed by ASTM E-8 shall be taken longitudinally from opposite sides of the body one inch from the front end. For the base plug, one tension test specimen conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the center of the plug transverse to the centerline of the part and parallel to the grain flow to the greatest extent possible. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or government from testing specimens at other locations to determine that the machenical properties of the body meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor the entire lot may be reheat treated and retested; or a new set of bodies or bases may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies or bases falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be reheat treated and retested.

4.5.2 Rotating band.

4.5.2.1 Seating (non-destructive method). The samples provided by 4.4.3.2.1 shall be tested for the requirements in 3.5.3 utilizing the equipment and procedures prescribed in the Detriot Testing Machine Manual (see 6.10).

4.5.2.2 Seating (destructive method). The body assembly selected in accordance with 4.4.3.2.2 should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60° apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper

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body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band. The 6 results shall be totaled and the sum thus obtained will be divided by 6 to determine the average diametrical clearance.

4.5.2.3 Sidewall contact. Samples provided by 4.4.3.2.3 shall be visually examined for scale formation on the surface which was adjacent to the seat and for contact with the sidewalls of the band seat.

4.5.2.4 Mechanical properties. The sample provided by 4.4.3.2.4 shall have the rotating band removal and carefully flattened without hammering so as to hold additional cold working to a minimum. A test specimen conforming to a sheet type, 1/2 inch wide, rectangular tension test specimen as prescribed by ASTM E-8 shall be prepared and tested in accordance with the procedures prescribed by ASTM E-8.

4.5.3 Protective coating. A salt spray test shall be performed on the projectile body and base plug utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable drawings and specifications. Failure to comply will be cause for rejection of the production represented by the sample.

4.5.3.1 Projectile body. Salt spray testing shall be conducted on production projectile bodies after the protective coatings have been applied (see dwg. 9362659). Sampling shall be in accordance with TT-C-490 when on reduced sampling (twice per week) one sample shall be taken during the first 30 minutes of production and the second taken during the last day of production that week.

4.5.3.2 Base plug. Salt spray testing shall be performed on production base plugs after the protective coating has been applied (see dwg. 9214146). Sampling shall be as specified in TT-C-490. When on reduced sampling (twice per week), one sample during the first 30 minutes of production and the second taken during the last day of production that week.

4.5.4 Moisture content of the obturating band. The obturating bands, after weighing on an analytic balance, shall be heated in an

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circulating oven for 24 hours at a temperature of 105° Centigrade (C). The bands shall be cooled to room temperature in a dessicator and reweighed. The bands shall be heated for one hour at 105°C, cooled and again reweighed. The procedure of heating for one hour, cooling and weighing shall be continued until the bands reach a constant weight. The percent moisture shall be calculated as follows:

Percent Moisture = Initial weight minus final constant weight divided by initial weight times 100

4.5.5 Obturator band strength. A steel fixture shall be employed which consists of a 5 inch diameter groove machined into two equal pieces of a rectangular steel plate. The groove shall be 5/16 inch wide and .400 + .030 inches deep. The fixture shall be placed to form a rounded groove disc and the band fitted in the machines groove. The steel plates shall then be attached to the self-aligning grips of a tensile testing machine by means of two suitable harnesses. Bands to be tested shall have a set direction of pull. The tensile testing machine shall be operated at a constant rate of 2.0 inches per minute until the band breaks. The pull experienced at break shall be recorded, and the elongation shall be calculated as follows. The elongation gage shall be adjusted to a zero setting after three pounds of pull is exerted on the band. After break, the elongation shall be calculated by the following formula. The temperature of the bands just prior to the test shall be between 65°F and 80°F.

$$\text{Percent Elongation} = \frac{2 \times \text{Elongation Reading}}{\text{Original circumference of band based on the inside diameter to nearest .1 inch}}$$

4.5.6 Magnetic particle inspection.

4.5.6.1 Projectile body. Heat treated projectile bodies after final machining but prior to banding shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method with indirect magnetization by either the continuous method or the residual method. The entire body inside and outside shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. The magnetic particle standard per dwg. 9280363 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.6 shall be rejected. Projectile bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

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4.5.6.2 Base plug. Base plugs after final machining shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method with indirect magnetization by either the continuous or residual methods. The magnetic particle standard per dwg. 9280364 shall be utilized to establish inspection sensitivity. Any base plug which fails to comply with the requirements of 3.6 shall be rejected. Base plugs which comply shall be stamped with the letter "M" as specified on the applicable drawing.

4.5.6.3 Suspension. The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867.

4.5.7 Proving ground ballistic tests. The following tests shall be conducted using cannon tubes having 25 percent minimum remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exits the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.9. Range, deflection, and velocity shall be recorded for each round fired for information.

4.5.7.1 Excess pressure. Each projectile submitted for this test shall be assembled with an inert loaded candle, a dummy fuze and an inert expelling charge. Each projectile will be fired from an M199 cannon tube with an M119A1 propelling charge weight adjusted and temperature pre-conditioned at +145°F to yield a chamber pressure of 115 + 2% of peak pressure. An attempt will be made to recover the projectiles fired for this test.

4.5.7.2 Service pressure. Each projectile submitted for this test shall be assembled with an inert loaded candle, an M565 Fuze and an expelling charge. Each projectile shall be fired for expulsion and recovery from an M185 of M199 cannon tube with an M119 series propelling charge.

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4.5.7.3 Sampling. Normal sampling shall be conducted until 3 consecutive lots have been accepted. Reduced sampling will then be instituted. Any failure occurring while on reduced sampling requires a return to normal sampling. Sample quantities for normal and reduced sampling shall be as follows:

	Normal	Reduced
a. Excess pressure	5	2
b. Service pressure	3	2

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Domestic shipment. Packaging, packing and marking shall be in accordance with the requirements of MIL-STD-1169, Level B and the following special instructions:

5.1.2 Obturating band packaging. Twenty-five (25) bands shall be placed in a paperboard box conforming to PPP-B-566, Variety 1, Style II, Type D, Class a. Box size shall be approximately 6 5/16 square by 9 3/8 inches long, inside dimensions with end opening. Eighteen (18) boxes (450 bands) shall be packed in a fiber box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 200 in 3 layers, 6 per layer, arranged 3 in width by 2 in length. A plain fiberboard pad of the same test board as the box shall be placed between layers, box size shall be approximately 19 1/8 by 19 1/8 inside dimensions. Pad size shall be equal to the inside length and width of the box less approximately 1/16 inch. Box closure shall be application of PPP-T-45, Type III, Grade B tape, 2 inch width over the entire length of all top and bottom seams with approximately 3 inch extension of tape around corners and onto the end panels.

5.2 Packing.

5.2.1 Domestic shipment.

5.2.1.1 Projectile body. The projectile bodies shall be packed in accordance with dwg. 7549275.

5.2.1.2 Base plug. The base plug shall be packed in accordance with Standard MIL-STD-1169, Level B.

5.2.1.3 Obturating band (if shipped separately). The obturating bands shall be packed in accordance with Standard MIL-STD-1169, Level B.

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5.3 Marking. Mark projectile bodies in accordance with dwg. 7549275 and base plugs and obturating bands per MIL-STD-1169.

6. NOTES

6.1 Intended use. The projectile metal parts contained in this specification are intended for use with Projectile, 155MM, Illuminating: M485A2.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, USAMCCOM (ARDC), ATTN: DRSMC-QAR-I, Dover, NJ 07801.

6.4 Date requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Date Requirement	Appliable DID
3.2, 3.5.2, 3.10	Procedure Report	* DI-T-5372 (Tailored)
3.5.3, 3.5.4, 4.5.1.1, 4.5.1.2, 4.5.2.1(2), 4.5.2.3, 4.5.2.4	Quality Inspection Test Demonstration & Evaluation Report	DI-R-1724
4.5.3.1, 4.5.3.2, 4.5.4, 4.5.5, 4.5.6	Quality Inspection Report	DI-R-1721

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, USAMCCOM (ARDC), ATTN: DRSMC-QAR-Q, Dover, NJ 07801.

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6.6 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.7 Moisturizing of obturating bands. The loading, assembling and packing contractor shall moisturize the bands by boiling in water until 2.0% to 3.0% moisture by weight is attained prior to assembling to the projectile.

6.8 Band seating. The operating procedures for use of Detroit Testing Machine (prepared by Inspection Engineering Office, US Army, Frankford Arsenal) submit requests to Commander, USAMCCOM (ARDC), ATTN: DRSMC-QAR-I, Dover, NJ 07801.

6.9 Steel blooms, billets, slabs and bars.

6.9.1 Steel grade. It is the responsibility of the manufacturer of the projectile metal parts to select a steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machinist to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

6.9.2 Raw material. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade (selected by the projectile manufacturer) and complying with the specified ASTM designation and tests slected by the projectile manufacturer.

6.9.3 Forging quality processing techniques. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machinist have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

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6.10 Preproduction sample/hardness range (projectile bodies and base plugs). To minimize the need for reheat treating material or for having miscellaneous heat treatment lots it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill heat. When using a hardness range acceptance will still be in accordance with paragraph 4.5.1.2.

6.11 Ballistic testing. Any evidence of excessive deformation in any phase of the proving ground tests will result in an immediate investigation by AMCCOM personnel.

Custodian:
Army-AR

Preparing activity:
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(Project 1320-A745)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48043A (AR)	Projectile, 155MM, Illuminating M485A2, Metal Parts Assembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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